How-To Library

Informatica

Deploy Axon Data Governance 7.3.1 on Kubernetes using Axon Deployment Manager

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Abstract

This article describes how to install Axon Data Governance on Kubernetes from the Docker image using Informatica Axon Deployment Manager. This article also describes how you can use Informatica Axon Deployment Manager to manage an existing Axon Data Governance deployment on Kubernetes.

Supported Versions

- Axon Data Governance 7.3.1

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Overview

A Kubernetes engine is a cluster comprised of one or more master nodes and multiple worker nodes. The Kubernetes cluster master runs the Kubernetes control plane processes, including the Kubernetes API server, scheduler, and core resource controllers. The worker nodes run the workloads.

Use the Web-based interface in Informatica Axon Deployment Manager to quickly create standard Docker images and deploy using Kubernetes.

Docker is an open source platform that provides an isolated environment called containers to run applications. Docker allows independent containers to run within a single Linux instance. A Docker image is an executable package that can run an application, a code, run-time files, environment variables, or configuration files. A container is a run-time instance of an image.

Axon Deployment Manager Process Checklist

Perform the following tasks associated with the installation:

1. Verify the following prerequisites:
   a. Verify the system requirements.
   b. Acquire the Axon Deployment Manager and Axon installer `.tar` files.
   c. Verify the license key.
   d. Verify the configuration of the Kubernetes cluster.
2. Build the Linux Docker image for the Axon services.
3. Run the Docker image to deploy Axon using Kubernetes.
4. Manage Axon using Axon Deployment Manager to perform tasks such as reindexing the database and restarting the Axon services.

Before You Begin

Verify System Requirements

Verify that your environment meets the minimum system requirements for the installation process.

- Disk space of 50 GB in the Docker build directory.
- Disk space of 50 GB in the current working directory.
- Disk space of 50 GB in the Docker configuration directory.
- Supported image name or one of the following image names for the subscribed Linux operating system:
  - `registry.access.redhat.com/rhel7:7.8`
  - `centos:7`
- Configure external storage and associate storage class with external storage.
- Disk space of 10 GB required for external storage.
- Master node configuration: 4 cores of 3 GHz with 24GB RAM
- Worker node configuration: 2 modules of 8cores of 3GHz with 24 GB RAM
- Register and subscribe to the machine where you build the docker images.

Acquire the Axon Installation Files

To acquire the Axon installation files and license key, contact Informatica Global Customer Support online.
When you contact Informatica Global Customer Support, they initiate a shipping request. The shipping team sends the following files to you:

Axon 7.3.1 `.tar` files

You require the Axon 7.3.1 `.tar` files to install the product.

The shipping team provides you access to the location of the Axon installer `.tar` files. Copy the Axon installer `.tar` file to the machine where you want to build the image.

License key

Before you install the software, verify that you have the license key available for the Axon installer.

The product is provided under the Bring Your Own License (BYOL) model. Copy the license key file to a directory that is accessible to the user account that installs the product. You can use an existing Informatica license, or contact Informatica Global Customer Support if you do not have a license key or if you have an incremental license key.

Axon Deployment Manager `.zip` file

Download and extract the `axon_deployment_manager.zip` file to install Axon Deployment Manager and deploy Axon on Kubernetes.

You can download the Axon Deployment Manager from Marketplace, visit https://marketplace.informatica.com/listings/on-premise/solutions/informatica-axon-deployment-manager.html.

Verify Kubernetes Configuration

Complete the following tasks on the machine where you run Axon Deployment Manager:

1. Install `kubectl`.
2. Ensure that the machine where the Axon Deployment Manager runs, can communicate with the Kubernetes cluster through the `kubectl cluster-info` command. The command displays information about the master and services that run in the cluster.

   For more information, see the Kubernetes documentation.

Use Docker Hub

You can use the Docker Hub to push the built image to your Docker repositories and use the image during the deployment instead of building the Docker image on every worker node.

If you want to pull an image from the Docker Hub or private registry, you must create a Kubernetes secret with the name `regcred` on each worker node. Run the following command to create the Kubernetes secret:

```
kubectl -n <namespace> create secret docker-registry regcred --docker-server=https://
index.docker.io/v2/ --docker-username=<username> --docker-password=<password>
```

Build the Docker Image

Build a Docker image for Axon. After you build the Docker image, Axon Deployment Manager uses the Docker image to run the Kubernetes pods. Ensure that the Docker image is stored in the local host or Docker registry before you install Axon in Kubernetes with Axon Deployment Manager.

To build the Docker image, perform the following tasks:

1. Launch Axon Deployment Manager
2. Choose the deployment type.
3. Configure the Docker image.
4. Configure the Axon services.

Note: If you already have a built Docker image stored in the Docker registry, you can directly run the Docker image to deploy Axon on the Kubernetes.

Step 1: Launch Axon Deployment Manager

To launch Axon Deployment Manager, perform the following steps:

1. Run the following command: `sh startup.sh`.

   The `sh startup.sh` command starts the following Axon Deployment Manager services:

   - appcon_web
   - build_run_service
   - client
   - cloud_gateway
   - validator_service

   The default port for Axon Deployment Manager is a numeric value assigned randomly from 12100 to 12200. To specify a different port, add the required port number at the end of the startup command, as shown in the following example:

   ```
   sh startup.sh <required port number>
   ```

   Note: If Axon Deployment Manager is already running, do not start it again.
2. On the **Home** page, select **Kubernetes**.
3. To build the Docker image, select **Build Image**.

   The **Deployment type** page appears.

Step 2: Choose the Deployment Type

On the **Deployment Type** page, select the deployment type and host instances to build the Docker image.

1. On the **Deployment type** page, select the product to build the Docker image.

   Axon is selected by default.
2. Optionally, you can choose to configure SSH protocol for secure authentication to build the image on the remote host.
   - If you do not want to build the image on the remote host, go to step 3.
   - If you want to build the image on the remote host, set the authentication type to **Password** or **Key**.
     - For password type authentication, enter the host name, user name, password, and port to connect to the host machine.
       The SSH connection uses port 22 by default.
     - For key type authentication, enter the host name, user name, path of the host authentication key, and port number.
3. Click **Next**.

Step 3: Configure the Docker Image

On the **Configure Docker** image page, enter the following information for the `.tar` files location and the images:

1. Specify the base operating system to build the Docker image.
   - To build the Docker image on CentOS, select `centos:7`.

- To build the Docker image on RHEL, select `registry.access.redhat.com/rhel7` or `registry.access.redhat.com/rhel7:7.8`.
- To build the Docker image on a custom operating system, select `Custom`, and enter the name of the valid operating system.

2. Enter the Axon 7.3.1 `.tar` file paths on the machine where you want to build the image.

   Axon Deployment Manager uses the tar utility to extract the installer server files to a directory on the machine.

3. Enter the working directory to build the Docker image.

   Default is `/root`.

4. To perform the cluster configuration to create multiple pods, you can select the following check boxes:
   - Select **Configure PostgreSQL cluster** to configure the PostgreSQL cluster.
   - Select **Configure Elasticsearch cluster** to configure the Elasticsearch cluster.
5. Click **Next**.

   The **Configure Axon Services** page appears.

Step 4: Configure the Axon Services

On the **Configure Axon Services** page, specify the image name for Axon services that you want to include in the Docker image.

1. In the **Image Name** section, specify the name of the Docker image for the Axon services. You can specify a default image name for the required Axon services.
   For example, for the Notification service, you can specify the default image name `informaticaltd/axonk8s:notification` or a custom image name to build the Docker image.
2. Click **Build**.

   The process of building the image starts.

After the Docker image is built, you can view the status and name of the image, along with the tasks performed by the installer in the log summary.

Run the Docker Image to Deploy Axon on Kubernetes

Run the Docker image to deploy Axon Data Governance on Kubernetes.

To run the Docker image, perform the following tasks:

1. Launch Axon Deployment Manager
2. Choose the deployment type.
3. Provide Axon deployment details.
4. Configure the Axon services.

Step 1: Launch Axon Deployment Manager (Run Image)

To launch Axon Deployment Manager, perform the following steps:

1. To launch Axon Deployment Manager, run the following command: `sh startup.sh`.

The `sh startup.sh` command will start the following Axon Deployment Manager services:

- appcon_web
- build_run_service
- client
- cloud_gateway
- validator_service

The default port information for Axon Deployment Manager is a numeric value assigned randomly from 12100 to 12200. To specify a different port, add the required port number at the end of the startup command as shown in the following example:

```
sh startup.sh <required port number>
```

Note: If Axon Deployment Manager is already running, do not start it again.

2. On the **Home** page, select **Kubernetes**.
3. To run the Docker image, select **Run Image**.

   The **Deployment type** page appears.

Step 2: Choose the Deployment Type (Run Image)

On the **Deployment Type** page, select the deployment type and host instances to run the Docker image.

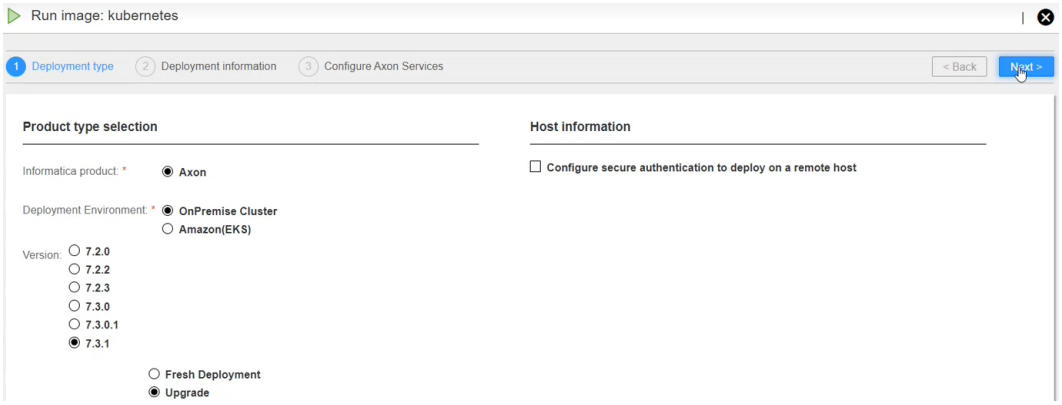
1. On the **Deployment type** page, select the product to build the Docker image.

   Axon is selected as the product by default.
2. In the **Deployment Environment** section, select any one of the following:
   - **OnPremise Cluster**: To deploy Axon on the onpremise cluster.
   - **Amazon(EKS)**: To deploy Axon on the Amazon Kubernetes cluster.

     Note: Before you deploy, ensure that you perform the following:

     - You should have Amazon Web Services account to create the Amazon (EKS) Kubernetes cluster.
     - Update Axon Docker images in the Docker hub.
3. Select **7.3.1** as the Axon version.
4. Select any one of the following options:
   - **Fresh Deployment**: To perform fresh deployment of Axon 7.3.1.
   - **Upgrade**: To upgrade from any previous version of Axon to Axon 7.3.1.

     Note: Ensure that you select **OnPremise Cluster** in the **Deployment Environment** section to upgrade to Axon 7.3.1.



5. Click **Next**.

   The **Deployment Information** page appears

6. In the **Working Directory** field, enter the path to the working directory.

   The working directory is the path to the previous Axon version.

7. Optionally, you can choose to configure the SSH protocol for secure authentication to deploy the containers on the remote host.

   - If you do not want to run the image on the remote host, go to step 6.
   - If you want to run the image on the remote host, set the authentication type to **Password** or **Key**.
     - For password type authentication, enter the host name, user name, password, and port to connect to the host machine.
       The SSH connection uses port 22 by default.
     - For key type authentication, enter the host name, user name, path of the host authentication key, and port number.

8. Click **Next**.

Step 3: Provide Axon Deployment Details

In the **Docker information** page, provide the required information based on the deployment type you selected to connect to the Docker image and container.

Fresh Deployment

If you select **Fresh Deployment** as the deployment type, execute the following steps.

1. In the **Docker and Kubernetes Information** section, configure the following fields:

Field	Description
Namespace	Enter the namespace of the host system where you want to install Axon. Enter lowercase alphanumeric characters and hyphen. The value must start and end with an alphanumeric character. Enter a maximum of 63 characters.
Storage Class	Specify storage class to manage external storage.
Informatica License Key	Enter the path and file name of the Informatica license key.

Field	Description
Working Directory	Enter the path to the directory where you want to install the utilities.

2. In the **Service configuration** section, configure the following options:
   - To configure the PostgreSQL database on an external deployment platform, such as Amazon RDS or Microsoft Azure Database, select **Use existing PostgreSQL database**, select the external deployment platform and enter the connection string.
     - To deploy the PostgreSQL database on the Amazon RDS instance, select **Amazon RDS** and specify the connection string in the following format:

       ```
       postgresql://db_user:db_password@hostname:db_port/<db_name>
       ```
     - To deploy the PostgreSQL database on the Microsoft Azure Database instance, select **Microsoft Azure Database** and specify the connection string in the following format:

       ```
       postgresql://db_user%40hostname:db_password@hostname:db_port/<db_name>?
       sslmode=require
       ```

     Note: When you configure the PostgreSQL database on external nodes such as Amazon RDS and Microsoft Azure Database, Informatica recommends that you ensure that the region of the external nodes is the same as that of the Kubernetes cluster nodes.
   - To view logs in the Admin Panel, select **Enable viewing of logs in the Admin Panel**.
   - For HTTP authentication, enter the port number in the range of 30000 to 32535.
     HTTP authentication uses port 31511 by default.
   - To configure SSL authentication, select **Enable HTTPS authentication**. Enter the port number, SSL certificate, and SSL key details.
   - To connect Axon to Enterprise Data Catalog, select **Connect Axon to Enterprise Data Catalog**. Enter the following details:
     - Enterprise Data Catalog Host
     - Enterprise Data Catalog Port
     - Enterprise Data Catalog Server Login User Name
     - Enterprise Data Catalog Server Password
     - Enterprise Data Catalog Server Namespace
     - Enterprise Data Catalog Resource Name
   - To connect Axon to Data Privacy Management, select **Connect Axon to Data Privacy Management**. Enter the following details:
     - Data Privacy Management Host
     - Data Privacy Management Port
     - Data Privacy Management Server Login User Name
     - Data Privacy Management Server Password
     - Data Privacy Management Server Namespace
   - To enable event monitoring, select **Do you want to enable Event Monitoring?**.
3. To perform the cluster configuration to create multiple pods, you can select the following check boxes:
   - Select **Configure PostgreSQL cluster** to configure the PostgreSQL cluster.
   - Select **Configure Elasticsearch cluster** to configure the Elasticsearch cluster.

You can enable auto-scaling for each of the above mentioned cluster configuration.

- Select **Enable Auto Scaling** check box to configure auto-scaling.
  Enter value as 2 or 3 for the minimum and maximum number of nodes. The default minimum value is 2 and maximum value is 3.
- select **Enable CPU Auto Scaling** check box to configure CPU auto-scaling.
  Enter value for **CPU Load Factor**. The default value is 75.

  If CPU utilization reaches CPU load factor, new pod gets created, load will be shared to newly created pod also.
- Select **Enable Memory Auto Scaling** check box to configure memory auto-scaling.
  Enter value for **Memory Loader Value**. The default value is 80.

  If memory utilization reaches memory loader value, new pod gets created, load will be shared to newly created pod also.

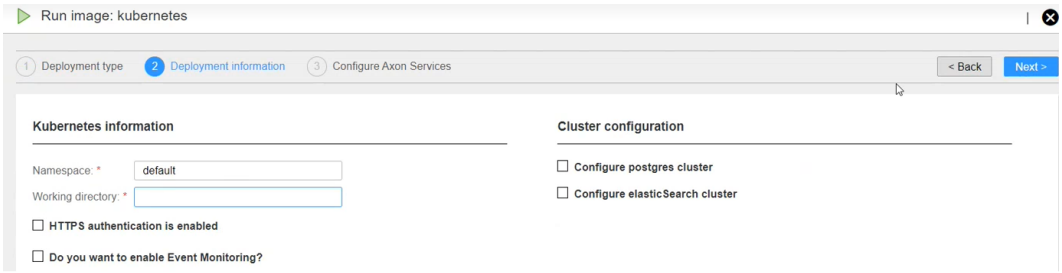
4. Click **Next**.

Upgrade to Latest Axon Version

If you select **Upgrade** as the deployment type, execute the following steps.

1. In the **Docker and Kubernetes Information** section, configure the following fields:

Field	Description
Namespace	Enter the namespace configured for the Axon 7.3.1 instance. Use lowercase alphanumeric characters and hyphen. The value must start and end with an alphanumeric character. Enter a maximum of 63 characters.
Working Directory	Enter the path to the existing Axon version directory where you want to deploy the latest Axon 7.3.1.



2. To deploy Axon 7.3.1 over previous Axon instance that has HTTPS authentication enabled, select the **HTTPS authentication is enabled** checkbox.
3. To enable event monitoring, select **Do you want to enable Event Monitoring?**.
4. To perform the cluster configuration to create multiple pods, you can select the following check boxes:
   - Select **Configure PostgreSQL cluster** to configure the PostgreSQL cluster
   - Select **Configure Elasticsearch cluster** to configure the Elasticsearch cluster.
5. Click **Next**.

Step 4: Configure the Axon Services (Run Image)

On the **Configure Axon Services** page, specify the image name for the Axon services that you want to deploy on Kubernetes.

1. In the **Image Name** section, specify the name of the Docker image for the Axon services. You can specify a default image name for the required Axon services.
   For example, for the Notification service, you can specify the default image name `informaticaltd/axonk8s:notification731` or a custom image name to build the Docker image.
2. Click **Deploy**.

   The Docker image runs and deploys Axon on Kubernetes.

In the log summary, you can view the status and details of the deployment logs along with the tasks performed by the installer.

Note: To access Axon, Informatica recommends that you use a fully qualified host name as specified in the following format: `http(s)://<full_qualified_hostname>:<port>/`

The `hostname` and `port` information are generated after you deploy Axon services on Kubernetes. We recommend that you do not you use the URL that is generated and displayed after the deployment is complete.

After you Deploy the Docker Image

After the Docker image is deployed, you must restart the bulk upload service and reindex the database. Additionally, if you plan to use the Axon instance with Enterprise Data Catalog, you must map the IP address and the domain name address of the Enterprise Data Catalog instance.

Reindex the Database

To ensure that segmentation is working as expected after you deploy Axon 7.3.1, reindex the database.

1. In Axon Deployment Manager, click **Manage Axon**.
2. In the **Action Information** section, enter the following details:

Field	Description
Namespace	Enter the same value that you used to run the Docker image to deploy pods in the Kubernetes environment.
Working directory	Specify the path to the existing working directory.
K8s Actions	Select **ReIndex**.

3. Click **Execute Action**.

Restart Bulk Upload Service

To ensure the bulk upload functionality is working as expected after you deploy the Docker image, restart the bulk upload service.

1. In Axon Deployment Manager, click **Manage Axon**.

2. In the **Action Information** section, enter the following details:

Field	Description
Namespace	Enter the same value that you used to run the Docker image to deploy pods in the Kubernetes environment.
Working directory	Specify the path to the existing working directory.
K8s Actions	Select **Restart the Axon services**. Under **Select the services to restart**, select the **bulkupload** micro service.

3. Click **Execute Action**.

Axon with Enterprise Data Catalog

To ensure the integration between Axon Data Governance and Enterprise Data Catalog is working as expected after you deploy the Docker image, map the IP address and the domain name address of the Enterprise Data Catalog host.

1. Update the `hosts` file in the Axon installation directory.
   a. Edit the `hosts` file using the following command:
      ```
      vi /etc/hosts
      ```
   b. At the end of the `hosts` file, add the IP address and domain name address of the Enterprise Data Catalog instance that you have integrated with Axon.
   c. Save the file.
2. Update the `hosts` file in the Kubernetes pod for Axon:
   a. Get the name of the Kubernetes pod for Axon using the the following command:
      ```
      kubectl get pods
      ```
   b. Open the Axon pod using the below command:
      ```
      kubectl exec -it <axon_pod_name> bash
      ```
   c. Edit the `hosts` file using the following command:
      ```
      vi /etc/hosts
      ```
   d. At the end of the `hosts` file, add the IP address and domain name address of the Enterprise Data Catalog instance that you have integrated with Axon.
   e. Save the file.

Manage Axon

You can restart Axon services using Axon Deployment Manager. You can take a backup of the database. You can also download logs to verify the status of your action.

1. To launch Axon Deployment Manager, run the following command: `sh startup.sh`

   Axon Deployment Manager uses port 12104 by default. To specify a different port, add the required port number at the end of the startup command as shown in the following example:

   ```
   sh startup.sh <required port number>
   ```

   Note: If Axon Deployment Manager is already running, do not start it again.
2. Click **Manage Axon**.

   Note: Steps 1 and 2 are not required if you are managing an existing deployment of Axon.

3. Optionally, if you want to remotely manage Axon that is deployed on Kubernetes, enable secure authentication, and set the authentication type to **Password** or **Key**.
   - For password type authentication, enter the host name, user name, password, and port to connect to the host machine.

     The SSH connection uses port 22 by default.
   - For key type authentication, enter the host name, user name, path of the host authentication key, and port number.
4. In the **Action Information** section, enter the following details:

Property	Description
Namespace	Enter the same value that you used to run the Docker image to deploy pods in the Kubernetes environment.
Working directory	Specify the path to the existing working directory.
K8s Actions	You can choose one of the following options: • Reindex the database. • Run the `paramsync` script. • Restart the Axon services. • Back up the database. • Check the status of the Axon services. • Configure services to create and refresh materialised view for Event Monitor service.

5. In the **Configure Services** section, select one of the following check boxes:
   - To create materialised view for Event Monitor service, select the **Do you want to enable matview?** check box.
   - To refresh the materialised view and view the latest data, select the **Do you want to refresh matview?** check box.

Frequently Asked Questions

Review the following frequently asked questions to understand the deployment of Axon on Kubernetes:

What happens if the Axon pod crashes or is deleted?

If the Axon pod crashes or is deleted, a new Axon pod is created and the old pod is terminated.

Where can I view the deployment logs?

You can view the logs in the working directory that you specified during the deployment of Axon on Kubernetes. The deployment log file name includes the latest timestamp. A sample deployment log file in the working directory is as follows:

```
/root/workingdirectory/logfile1619363942811.log
```

Where can I find the database backup?

You can access the database backup from the following location:

```
<Mount location for persistent volume>/<Shared volume pvc>/db_backup
```

A sample database backup directory is as follows:

```
/nfsshare/default-shared-volume-pvc-1510ff6b-8272-4334-9b7e-abaed1334720/db_backup
```

Is there a prerequisite for integration of products with Axon that is deployed on Kubernetes?

To integrate other products, such as Axon Agent, Enterprise Data Catalog, Informatica Data Quality, and Data Privacy Management, with Axon that is deployed on Kubernetes, add a DNS entry for the host name of the products. The server in which other products are installed can be resolved by using a fully qualified host name.

Which services are included in the Axon pod?

An Axon pod includes the following services:

- `Assetonboarder`
- `Ldapsynchronizer`
- `jobber`
- `daemontools`
- `apache`
- `odbconsumer`
- `php`
- `wkhtmltopdf`

How do I run the `php bin/console` commands in the Kubernetes environment?

To run the `php bin/console` commands, perform the following steps:

1. To log in to the Axon pod, run the following command:

   ```
   kubectl -n <namespace> exec -ti <axon_pod_name> /bin/bash
   ```

2. To change the user, run the following command:

   ```
   su nroot
   ```

3. Run the `php bin/console` commands by using the following syntax:

   ```
   export /home/nroot/exportEnv.sh;cd /home/nroot/axon;/home/nroot/php/bin/php bin/
   console <command>
   ```

Author